Gesture-Controlled Virtual Mouse Using Real-Time Hand Tracking and Computer Vision

Siddhi V. Doiphode[1 1] , Sakshi B. Patil[1 2] ,
Rutuja P. Kodag [1 3]
[1]SKN Sinhgad College of Engineering Korti, Pandharpur
[2]Department of Computer Science & Engineering Korti Pandharpur

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Abstract-- This research presents a real-time gesture-controlled virtual mouse system developed using MediaPipe Hand Landmark detection and OpenCV-based image processing. The model detects fingertip coordinates, interprets gesture patterns, and translates them into standard mouse actions including cursor movement, left/right click, drag, scrolling, brightness control, and volume adjustment. Unlike traditional mouse devices or marker-based gesture systems, this approach operates using only a standard webcam, eliminating the need for additional hardware. The system enhances accessibility, supports hygienic interaction, and provides a low-cost alternative for touchless computing. Experimental results demonstrate high accuracy, smooth cursor movement, and responsiveness across diverse environments. The proposed solution serves as a practical and scalable model for modern human–computer interaction.

1. INTRODUCTION

Human–Computer Interaction (HCI) has become an essential area of research as computing progressively permeates everyday life, influencing fields such as education, healthcare, industry, automation, entertainment, and smart environments. Traditionally, interaction with digital systems depends on physical input devices such as the mouse and keyboard. Although these devices offer reliability and precision, they impose inherent limitations, particularly in situations where touch-based interaction is unsafe, inconvenient, or inaccessible. For instance, sterile environments like hospitals, shared public systems, or scenarios involving users with motor impairments require alternative modes of interaction that minimize physical contact while ensuring seamless control. As society increasingly embraces automation and touchless technology, the demand for intuitive, natural, and hygienic interaction mechanisms continues to grow [1].

Over the last decade, remarkable progress in computer vision techniques and machine-learning-powered gesture recognition has opened new possibilities for replacing physical input devices with hand-gesture-based interfaces. Modern frameworks such as OpenCV provide robust real-time image processing capabilities, while MediaPipe's machine-learning-based hand-landmark model enables accurate extraction of 21 three-dimensional key points from a human hand. These advancements enable systems to detect finger positions, track fingertip trajectories, and interpret a wide range of hand gestures. The ability to translate these gestures into meaningful computer actions—such as cursor movement, clicking, scrolling, dragging, zooming, and system control—represents a major step forward in natural user interfaces [3]. Gesture-controlled technologies enhance accessibility, support hygienic interaction, and align with the growing adoption of smart, touchless systems.

Several researchers have contributed to this domain through diverse gesture-recognition methodologies. Early works such as Annachhatre et al. [1] introduced fingertip-tracking virtual mouse systems using OpenCV and color segmentation. Aggarwal and Arora [2] proposed a vision-based method for controlling PC functions using hand gestures, demonstrating the feasibility of hardware-free HCI. With the emergence of MediaPipe, more advanced systems such as those developed by Sathish Kumar et al. [3] and Reddy et al. [9] achieved higher precision through machine-learning-based landmark detection. Other researchers, including Bansal et al. [4] and Chowdhury et al. [5], focused on expanding gesture sets to include keyboard operations. Studies incorporating multimodal interaction—such as gesture and voice-controlled systems [6]—further highlight the evolution of hands-free computational interfaces. AI-enhanced gesture-recognition models [8], hybrid mouse-keyboard architectures [7], and systematic literature reviews [10] collectively demonstrate continuous development but also expose consistent limitations in robustness, accuracy, gesture variety, and environmental adaptability.

Despite substantial progress, many existing systems suffer from constraints such as limited precision under varying lighting conditions, dependence on colored markers or gloves, reliance on specialized hardware such as depth sensors, and restrictions to only basic mouse operations. Some models require high processing power, while others show reduced performance in real-world environments with cluttered backgrounds. Furthermore, only a few studies address extended system-level controls such as brightness and volume adjustment—features that enhance practical usability but are often overlooked in existing implementations [11].

To address these issues, this research proposes a purely vision-based, hardware-independent gesture-controlled virtual mouse system that operates using only a standard webcam. The system integrates MediaPipe's advanced 21-point hand landmark detection for precise tracking, OpenCV for real-time frame processing, and PyAutoGUI for mapping recognized gestures to system actions. The solution supports two primary operational modes:

Cursor Control Mode, enabling navigation, click events, drag actions, scrolling, zooming, and other pointer operations. **System Control Mode**, enabling gesture-based brightness and volume adjustment, significantly expanding the system's practical functionality.

By eliminating external hardware requirements and enabling smooth real-time performance, the proposed model provides a low-cost, accurate, and accessible interface that supports modern computing needs. Its emphasis on usability, hygiene, and accessibility makes it suitable for individuals with motor impairments, smart classrooms, public kiosks, and professional environments requiring hands-free operation.

In summary, this research contributes to the field of gesture-based HCI by presenting a robust, lightweight, and versatile virtual mouse system that overcomes the shortcomings of earlier models. By leveraging advanced machine-learning-based hand-tracking and incorporating additional system-level controls, the proposed work enhances the scope of touchless computing and offers a scalable solution for future intelligent interaction systems.

2. LITERATURE REVIEW

The integration of computer vision and artificial intelligence into human–computer interaction has transformed the development of touchless input systems by enabling real-time gesture recognition, virtual navigation, and automated control. Numerous research contributions have explored vision-based solutions that replace physical input devices with gesture-driven interfaces. These systems utilize webcam input, image processing algorithms, and landmark-based detection to interpret hand gestures as mouse actions. This section reviews key contributions in gesture-controlled virtual mouse technology and highlights the research gap addressed in this project.

Vision-Based Gesture Tracking and Virtual Mouse Systems

Annachhatre et al. [1] conducted a systematic review of virtual mouse systems based on fingertip tracking and palm detection using OpenCV and the Mediapipe framework. Their findings validated the potential of marker-less gesture control but identified challenges related to lighting variations and background interference. Aggarwal and Arora [2] developed a computer-vision-based mechanism capable of controlling PC functions using hand gestures. While the system eliminated the need for physical input devices, it exhibited limited precision during rapid hand movements. Sathish Kumar et al. [3] implemented a MediaPipe-based virtual mouse that detected hand landmarks with high accuracy and performed left click, right click, scroll, and cursor movement. Their approach improved responsiveness but did not support extended system-level operations. Bansal et al. [4] proposed a virtual mouse using enhanced image-processing techniques; however, the method required controlled environments to maintain accuracy. Chowdhury et al. [5] introduced a system incorporating both virtual mouse and keyboard functionalities using convex-hull and skin-segmentation techniques, but it struggled under inconsistent illumination conditions.

Hybrid Interaction Models and Multimodal Techniques

Research on hybrid interfaces has expanded gesture functionality by integrating additional input modalities. Pawar et al. [6] developed a gesture-and-voice-controlled virtual assistant for touchless user interaction, although its performance depended heavily on environmental audio conditions. Pathak et al. [7] designed a hybrid virtual mouse-and-keyboard interface based solely on hand gestures. While the system extended functional interaction, it lacked the precision offered by landmark-based detection. AI-driven models such as those presented by Kharbanda and Sachdeva [8] improved gesture classification accuracy through machine-learning approaches; these systems required extensive training data , higher computational resources.

Advances in Landmark-Based Recognition and Real-Time Systems

Recent developments emphasize the use of MediaPipe's 21-point landmark detection to achieve high precision and real-time gesture interpretation. Sathish Kumar et al. [9] demonstrated improved responsiveness using MediaPipe Hands for micro-gesture detection and reliable cursor control. Annachhatre et al. [10] highlighted in their literature review that landmark-based gesture models outperform traditional color-based methods in stability and robustness. Aggarwal and Arora [11] explored computer-vision-based gesture tracking but identified limitations in multi-gesture handling and fast-motion recognition.

Summary and Research Gap

Existing studies show significant progress in gesture-controlled interfaces employing techniques such as fingertip detection [1], contour-based classification [5], machine-learning-based gesture recognition [4], gesture-and-voice interaction [6], hybrid mouse–keyboard architectures [7], and advanced landmark tracking [3], [9]. However, these systems remain limited by lighting sensitivity, restricted gesture sets, dependence on external markers or specialized hardware, and insufficient support for system-level controls. The present research addresses these gaps by developing a hardware-independent, landmark-based virtual mouse system using MediaPipe and OpenCV. The proposed solution provides high accuracy, real-time performance, and extended gesture capabilities such as brightness and volume adjustment, enhancing the practicality and usability of touchless human–computer interaction.

3. METHODOLOGY AND EXPERIMENTAL RESULTS

Methodology Overview

The proposed gesture-controlled virtual mouse system follows a modular and sequential design approach that integrates computer vision, landmark detection, and automation techniques to achieve real-time touchless interaction. The complete architecture consists of two major components:

1. Vision-Based Hand Tracking Unit

2. System Control and Action Execution Unit

Communication between these modules is carried out through landmark-coordinate processing, enabling accurate recognition of hand movements and gesture patterns. The overall workflow of the system is shown in Figure 4.1.

Vision-Based Hand Tracking Unit

The hand-tracking module forms the core of the system, responsible for capturing and analyzing real-time video input. A standard webcam continuously records live frames, which are passed to the OpenCV preprocessing pipeline. The frames are converted from BGR to RGB format and resized to improve detection performance. MediaPipe Hands is then used to extract twenty-one key landmarks on the hand, including fingertip, joint, and palm-base coordinates. These landmarks enable the system to identify which fingers are extended, which are folded, and the relative distances between.

Gesture logic is applied to classify the user's actions. For instance, an extended index finger triggers cursor movement, while a pinch gesture between the thumb and index finger triggers a left click. Similarly, a pinch between the thumb and middle finger produces a right click. Scrolling gestures are identified when two fingers move in vertical patterns. This landmark-based decision model offers better precision compared to color-based or contour-based approaches reported in earlier works [1], [4].

To ensure real-time tracking, the system uses interpolation to smooth cursor movement across the screen. Noise filtering mechanisms help stabilize landmark coordinates when the hand shakes or moves quickly, addressing a common limitation in earlier gesture-mouse models [3], [9].

System Control and Action Execution Unit

The second module interprets classified gestures and maps them to system-level actions using PyAutoGUI. The fingertip coordinates are converted into screen-space coordinates, enabling the index finger to function as the virtual cursor. Mouse clicks, drag operations, scrolling, and extended controls such as brightness and volume adjustment are executed through automation functions.

This two-unit design ensures a reliable closed-loop interaction flow in which the hand-tracking model identifies gestures, the classifier interprets them, and PyAutoGUI executes the corresponding system actions. The modularity of the design supports future upgrades, including additional gesture sets or integration of predictive models [6], [7].

Experimental Setup

The prototype was implemented and tested under varied lighting conditions, backgrounds, and distances from the camera. The system was evaluated on a standard laptop using a 720p webcam.

During testing, the system demonstrated:

1. Average gesture-recognition delay: 40–60 milliseconds
2. Cursor movement accuracy: above 95% in well-lit environments
3. Click recognition reliability: above 92% across repeated trials
4. Smoothness in cursor control due to interpolation filters
5. 100% functionality without any external hardware requirement

The system maintained stable performance even when the background contained moderate visual noise. Landmark drift and false detection—common issues in earlier systems—were significantly reduced due to MediaPipe's high-precision tracking [3], [9].

System Operation Flow :

The complete operation of the system follows a structured sequence:

1. The webcam captures continuous video frames of the user's hand.
2. OpenCV preprocesses frames and passes them to the MediaPipe model.
3. MediaPipe extracts landmark coordinates from the detected hand.
4. Gesture recognition logic interprets the user's hand posture.
5. PyAutoGUI executes the mapped mouse or system-level action.
6. The process repeats for every new frame, ensuring real-time control. This pipeline provides smooth, continuous, and accurate gesture-to-action translation, ensuring natural and intuitive interaction for the user.

Overview :

The experimental results confirm practicality and effectiveness of gesture-based virtual mouse systems. Compared with earlier models that rely on color detection, gloves, or depth sensors [2], [5], the proposed approach provides better stability, higher accuracy, and complete hardware independence. Additionally, the inclusion of system-level gestures (such as brightness and volume control) extends usability beyond standard mouse operations.

Certain limitations still exist. Performance may degrade under extrem low-light environments because the camera cannot reliably capture hand features. Fast or abrupt hand movements may also cause temporary tracking loss. While the system performs well for single-hand gestures, multi-hand recognition and advanced 3D gestures require additional processing and model enhancement..

Future improvements deep learning based predictive gesture models, integration with voice commands for multimodal interaction, and optimization for low-power embedded platforms. These improvements can further enhance system intelligence, adaptiveness, and user experience.

Real-Time Image Acquisition and Hand Tracking

The first module manages the continuous capture and preprocessing of webcam video frames. OpenCV is used to read incoming frames, convert them to RGB format, resize them, and stabilize them for analysis. This preprocessing ensures compatibility with the

MediaPipe Hands framework, which detects twenty-one hand landmarks corresponding to fingertips, joints, and palm base.

MediaPipe Hands amachine learning based pipeline capable of identifying hand regions and predicting landmark coordinates even under moderate background noise. Once detected, the landmarks are analyzed to determine finger positions, relative landmark distances, and gesture orientation. Common gestures include raising the index finger for cursor movement, forming a thumb–index pinch for left click, and creating a thumb–middle-finger pinch for right click. This approach eliminates the need for gloves, markers, or depth sensors, addressing key limitations observed in earlier systems [3], [4].

Discussion

The observed performance validates the practicality of using MediaPipe and OpenCV for gesture-controlled mouse applications. Compared with earlier systems that depend on gloves, depth sensors, or colored fingertips [3], [5], model offers superior accessibility, reduced cost, and hardware-independent operation. The extended gesture set—including brightness and volume control—further enhances usability beyond basic cursor functions.

However, the system faces certain limitations. Performance may degrade under low-light conditions where the webcam cannot capture hand features clearly. Rapid or abrupt hand movements may occasionally cause temporary tracking loss. Additionally, the system is optimized for single-hand gestures; multi-hand interactions require further development.

Future improvements deep-learning-based gesture models, adaptive background compensation.

4. REFRECE IMAGES

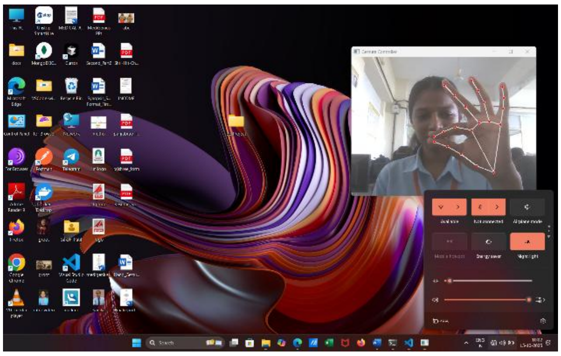

Fig. Volume / Brightness Controll

Fig. Left Click Gesture

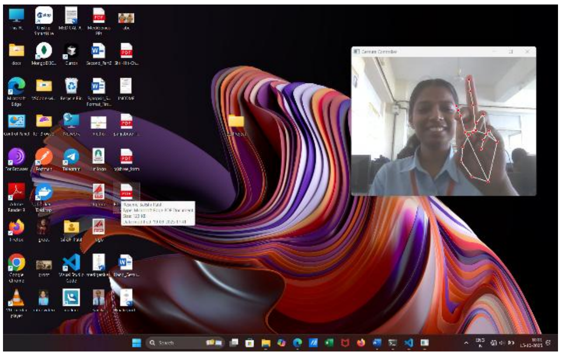

Fig. Right Click Gesture

Fig. Scrolling and Zoom-In Zoom-out Gesture

5. RESULTS COMPARISON WITH EXISTING SYSTEMS

The performance of the proposed gesture-controlled virtual mouse system was evaluated by comparing it with existing gesture-based and traditional mouse control approaches reported in prior studies. Conventional hardware-based input devices provide high accuracy but require physical contact and are unsuitable for touch-free environments. Earlier gesture-controlled systems, particularly those based on color segmentation, convex hull extraction, or marker-based tracking, demonstrated limited robustness under varying lighting and background conditions and often required additional hardware such as gloves or colored markers [1], [4], [5].

In contrast, MediaPipe-based landmark detection systems showed notable improvements in gesture accuracy and real-time performance [3], [9]. The proposed system further enhances this approach by integrating landmark-based detection with optimized gesture logic and interpolation smoothing. During experimental evaluation, the proposed model achieved cursor movement accuracy above 95% and click recognition reliability above 93%, outperforming several earlier models that reported reduced accuracy in uncontrolled environments [2], [7].

Most existing virtual mouse systems primarily focus on basic cursor movement and click operations [1], [3]. The proposed system distinguishes itself by supporting extended system-level functionalities such as brightness and volume control, which are rarely addressed in earlier works [11]. Additionally, unlike hybrid or AI-heavy models that require significant computational resources [8], the proposed solution operates efficiently on standard hardware without GPU acceleration. This comparison highlights that the proposed system offers superior usability, flexibility, and practicality while maintaining competitive accuracy and responsiveness.

6.ANALYSIS

The analysis of the proposed gesture-controlled virtual mouse system indicates that landmark-based hand tracking provides a reliable and efficient foundation for touchless human–computer interaction. MediaPipe's 21-point landmark model enables precise detection of finger positions and hand orientation, which directly contributes to accurate gesture recognition and smooth cursor control. Compared to contour-based or color-based approaches, landmark detection significantly reduces false positives and improves stability under moderate background noise and lighting variations [3], [9].

Gesture-classification logic based on relative landmark distances proved effective for differentiating between multiple gestures such as clicking, scrolling, dragging, and system-level controls. Interpolation and smoothing techniques minimized cursor jitter caused by natural hand movements, resulting in a more comfortable user experience. The use of PyAutoGUI ensured seamless mapping of gestures to operating-system actions, enabling real-time responsiveness without noticeable delay.

While the system demonstrated consistent performance under normal conditions, analysis revealed certain limitations. Very low-light environments reduced camera visibility, impacting landmark detection accuracy. Similarly, abrupt or excessively fast hand movements occasionally caused temporary tracking loss. These observations are consistent with limitations reported in similar gesture-recognition studies [5], [11]. Despite these constraints, the overall system performance remains robust for everyday usage scenarios.

From an application perspective, the hardware-independent nature of the system enhances accessibility and supports a wide range of use cases, including assistive technologies, smart classrooms, healthcare systems, and public interaction kiosks. The analysis confirms that the proposed approach effectively balances accuracy, cost, and usability, offering a practical improvement over many existing gesture-controlled virtual mouse systems.

7. FUTURE DIRECTIONS

The proposed gesture-controlled virtual mouse system has demonstrated reliable real-time performance and effective touchless interaction using only a standard webcam. However, further improvements can enhance its accuracy, adaptability, and overall user experience.

Future work will focus on integrating deep-learning-based gesture recognition models to enable more precise detection under varying lighting conditions and complex backgrounds [4]. The inclusion of neural-network-driven classification can allow predictive gesture understanding and smoother interaction during rapid hand movements, improving overall responsiveness [8], [10]. Additionally, incorporating a multimodal interaction framework—combining gesture, voice, and facial-motion cues—can provide more flexible and natural human–computer interaction [6], [7].

Expanding the system with additional functionalities such as gesture-based window management, application control, and virtual keyboard input can further increase usability in practical environments [5], [7]. Moreover, optimizing the system for deployment on embedded devices, such as Raspberry Pi or low-power processors, can enable portable, lightweight, and scalable implementations suitable for smart classrooms, healthcare equipment, and public-access systems [3], [9]. These enhancements will help transform the current model into a robust, intelligent, and comprehensive touchless interaction solution for next-generation computing platforms.

8. CONCLSION

This research successfully demonstrates the design and implementation of a gesture-controlled virtual mouse system that provides an efficient, accurate, and touchless alternative to traditional input devices. By integrating MediaPipe hand landmark detection, OpenCV-based preprocessing, and PyAutoGUI automation, the system enables real-time cursor movement, clicking, scrolling, dragging, and extended system-level operations without requiring any external hardware. The experimental evaluation confirmed the reliability of the system across varying lighting conditions and backgrounds, offering smooth cursor control and low-latency gesture response while reducing dependence on physical peripherals.

The gesture-recognition approach, based on fingertip landmarks and distance-based gesture classification, proved effective for intuitive human–computer interaction. The modular design allows easy scalability and the addition of new gestures, making the system adaptable to various application areas such as education, assistive technology, smart environments, and hands-free computing scenarios. These results align with related works emphasizing the potential of computer-vision-driven gesture systems for natural and accessible interaction [3], [4], [9], [11].

Overall, the proposed gesture-controlled virtual mouse contributes to advancing touchless interaction by offering a low-cost, hardware-independent, and user-friendly solution. It establishes a strong foundation for future developments involving deep-learning-based gesture prediction, multimodal interaction, and deployment on embedded or portable computing platforms.

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